

Margin of Error

Margin of Error tells us how much our sample estimate may differ from the actual population value.

In simple words:

It tells us the “plus or minus” amount around our sample mean.

Example

Suppose:

Sample Mean=70

And:

Margin of Error=4

Then:

Confidence Interval= 70 ± 4

So:

(66, 74)

How do we calculate it?

For the Z-distribution:

$$ME = z^* \times SE$$

$$SE = \frac{\sigma}{\sqrt{n}}$$

For the t-distribution:

$$ME = t^* \times SE$$

$$SE = \frac{s}{\sqrt{n}}$$

Computing CI: z-score method

The main idea

We start with a sample mean.

For example:

We want to estimate the average height of all students, but we only collect data from a sample.

We calculate the sample mean:

$$\bar{x}$$

But the sample mean may not exactly equal the true population mean.

So, we create a range:

$$\text{Confidence Interval} = \text{Sample Mean} \pm \text{Margin of Error}$$

Main formula

For the Z-score method:

$$CI = \bar{x} \pm Z * \frac{\sigma}{\sqrt{n}}$$

This can be broken into three parts:

Sample Mean

$$\bar{x}$$

Standard Error

$$SE = \frac{\sigma}{\sqrt{n}}$$

Margin of Error

$$ME = z \times SE$$

Finally:


$$CI = \bar{x} \pm ME$$

Critical Z-value for a 95% confidence level

Step 1: Start with the confidence level

$$95\% = 0.95$$

Step 2: Find the area outside the confidence interval

$$1 - 0.95 = 0.05$$

So, 5% is outside the middle 95%.

Step 3: Divide the remaining area between two tails

$$\frac{0.05}{2} = 0.025$$

So, each tail contains:

0.025

Step 4: Find the area to the left of the positive critical Z-value

Most common Z-tables show the cumulative area from the **far left**.

Therefore:

$$1 - 0.025 = 0.975$$

So, look for: **0.975**

in the Z-table.

Step 5: Read the corresponding Z-value

In the Z-table:

- Row: **1.9**
- Column: **0.06**

Therefore:

$$Z = 1.96$$

So, for a **95% two-tailed confidence interval**:

$Z_{\text{critical}} = \pm 1.96$

For the confidence interval formula, we generally use the positive value: **1.96**

Confidence Level	Critical Z-value
90%	1.645
95%	1.96
99%	2.576

Example: Average Height of Students

Suppose you want to know the **average height of all students** in a college.

You cannot measure every student, so you select a sample of **100 students**.

You find:

- Average height of the sample = **170 cm**
- Population standard deviation = **10 cm**
- Sample size = **100**
- Confidence level = **95%**

Our goal is:

Find the likely range of the true average height of all students.

Step 1: Start with the sample mean

Our best estimate of the population mean is:

$$\bar{x} = 170$$

So, we start at: 170 cm

But the actual population average may be slightly different.

Therefore, we need a range around 170.

Step 2: Find the Z-value

For a **95% confidence level**:

$$Z = 1.96$$

This means we want to get approximately **1.96 standard errors on either side** of our sample mean.

Step 3: Calculate the Standard Error

The formula is:


$$SE = \frac{\sigma}{\sqrt{n}}$$

Substitute the values:

$$SE = \frac{10}{\sqrt{100}} = 1$$

So:

$$SE = 1$$

Step 4: Calculate the Margin of Error

The formula is:

$$ME = z \times SE$$

$$ME = 1.96 \times 1 = 1.96$$

This means our sample mean may vary by approximately **1.96 cm**.

Calculate the Confidence Interval

The formula is:


$$\text{Confidence Interval} = \text{Sample Mean} \pm \text{Margin of Error}$$

So:

$$\text{CI} = 170 \pm 1.96$$

Lower limit:

$$170 - 1.96 = 168.04$$

Upper limit:

$$170 + 1.96 = 171.96$$

Therefore:

$$\text{95\% CI} = (168.04, 171.96)$$

In simple words

We took a sample of 100 students and found their average height to be 170 cm.

Using the Z-score method, we estimate that the actual average height of all students is likely between:

168.04 cm and 171.96 cm at a **95%** confidence level.

ACTIVITY

A company wants to estimate the **average daily screen time** of its employees.

A random sample of **100 employees** is selected.

The following information is available:

- Sample mean ($\bar{x}$) = **6**
- Population standard deviation (σ) = **2 hours**
- Sample size (n) = **100**
- Confidence level = **95%**

Task:

Calculate the **90% confidence interval** for the true average daily screen time.

What do you think will happen to the confidence interval if we increase the confidence level from 90% to 99%?

t-Distribution Method

The **t-distribution method** is very similar to the **Z-score method**.

The main difference is:

Z-distribution:

Use when the **population standard deviation σ** is known.

$$CI = \bar{x} \pm Z * \frac{\sigma}{\sqrt{n}}$$

t-distribution:

Use when the **population standard deviation is unknown** and we use the **sample standard deviation s** .

$$CI = \bar{x} \pm t * \frac{s}{\sqrt{n}}$$

Simple Example

Suppose we want to estimate the average test score of students.

We have:

- Sample mean:

$$\bar{x} = 70$$

- Sample standard deviation:

$$s = 10$$

- Sample size:

$n=25$

- Confidence level:

95%

Step 1: Calculate Degrees of Freedom

Degrees of freedom means the number of values that are free to vary in a calculation.

Degrees of freedom tells us how much independent information is available for estimating variation. For a one-sample t-test or t-confidence interval, we lose one degree of freedom because the sample mean is estimated from the same data, so $df=n-1$.

For the t-distribution:

$$df = n - 1$$

$$df = 25 - 1$$

$$df = 24$$

Step 2: Find the Critical t-value

For:

- Confidence level = 95%
- Degrees of freedom = 24

For a 95% confidence interval:

95% is in the middle

The remaining:

$$100\% - 95\% = 5\%$$

is divided into two tails:

2.5%

Use the t-table

Now we have two pieces of information:

Degrees of freedom:

df=24

Tail probability:

0.0250

Now go to the **t-table**.

- Find the row for $df=24$
- Find the column for a two-tailed 95% confidence interval, usually labelled:
 - **0.025 one-tail**, or
 - **0.05 two-tail**

At their intersection, you get:

$$t = 2.064$$

The t-value depends on both the confidence level and the degrees of freedom.

Step 3: Calculate Standard Error

$$SE = \frac{s}{\sqrt{n}} = \frac{10}{\sqrt{25}} = 2$$

Step 4: Calculate Margin of Error

$$\begin{aligned}
 ME &= t \times SE \\
 &= 2.064 \times 2 \\
 &= 4.128
 \end{aligned}$$

Calculate Confidence Interval

$$CI = \bar{x} \pm ME$$

$$CI = 70 \pm 4.128$$

Lower limit:

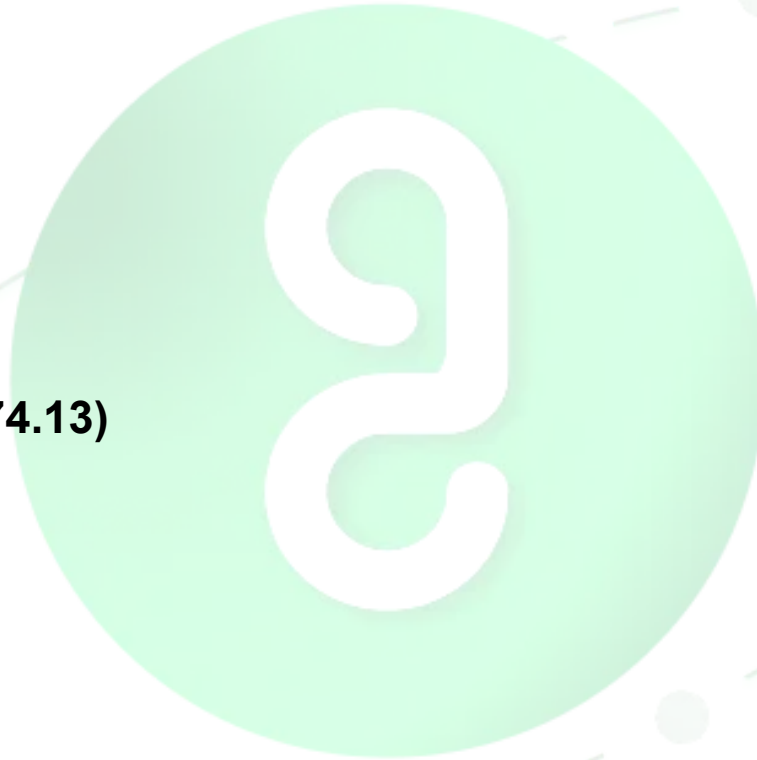
$$70 - 4.128 = 65.872$$

Upper limit:

$$70 + 4.128 = 74.128$$

Therefore:

$$\mathbf{95\% \text{ CI} = (65.87, 74.13)}$$



A teacher wants to estimate the **average marks of all students** in a college.

A random sample of **25 students** is selected.

The sample results are:

- Sample mean $\bar{x}=70$
- Sample standard deviation (s)=10
- Sample size (n)=25
- Confidence level = **95%**

Calculate

1. **Degrees of Freedom**
2. **Critical t-value**
3. **Standard Error**
4. **Margin of Error**
5. **95% Confidence Interval**
6. **Interpret the result**